

T3 UV and EFT Lagrangians

EFT order: 5; loop order: 1.

$$T3 - B, \quad \alpha = -1 : \quad S_1 = (2, -\frac{1}{2}), \quad S_2 = (2, \frac{1}{2}), \quad F = (1, 0)$$

$$\begin{aligned}
& \textbf{Full UV Lagrangian} - \frac{B_{\mu_1 \mu_2}^2}{4g_Y^2} - \frac{\left(G_{\mu_1 \mu_2}^{X_1}\right)^2}{4g_s^2} \\
& - \frac{\left(W_{\mu_1 \mu_2}^{A_1}\right)^2}{4g_L^2} \\
& + i \overline{d_{\alpha_1, i_1}} \gamma^{\mu_1} P_R D_{\mu_1} (d_{\alpha_1, i_1}) \\
& + i \overline{e_{i_1}} \gamma^{\mu_1} P_R D_{\mu_1} (e_{i_1}) \\
& + i \overline{q_{\alpha_1, a_1, i_1}} \gamma^{\mu_1} P_L D_{\mu_1} (q_{\alpha_1, a_1, i_1}) \\
& + i \overline{u_{\alpha_1, i_1}} \gamma^{\mu_1} P_R D_{\mu_1} (u_{\alpha_1, i_1}) \\
& + i \overline{\ell_{a_1, i_1}} \gamma^{\mu_1} P_L D_{\mu_1} (\ell_{a_1, i_1}) \\
& + \frac{1}{2} i N_{r_1}^T C \gamma^{\mu_1} D_{\mu_1} (N_{r_1}) \\
& - \frac{1}{2} N_{r_1}^T C N_{r_1} M_F \\
& + \overline{\ell_{a_1, i_1}} P_R N_{r_1} \text{NewScalar1}_{a_1} y_{1, i_1, r_1} \\
& - \overline{q_{\alpha_1, a_1, i_1}} P_R d_{\alpha_1, i_2} H_{a_1} Y_{d, i_1, i_2} \\
& - \overline{\ell_{a_1, i_1}} P_R e_{i_2} H_{a_1} Y_{e, i_1, i_2} \\
& - \overline{Y_{u, i_1, i_2}} \overline{u_{\alpha_1, i_2}} P_L q_{\alpha_1, a_1, i_1} H_{a_2} \bar{e}_{a_1, a_2} \\
& + \overline{y_{2, i_1, r_1}} N_{r_1}^T C P_L \ell_{a_1, i_1} \text{NewScalar2}_{a_2} \bar{e}_{a_1, a_2} \\
& - \overline{Y_{d, i_1, i_2}} \overline{d_{\alpha_1, i_2}} P_L q_{\alpha_1, a_1, i_1} H_{a_1}^\dagger \\
& - \overline{Y_{e, i_1, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} H_{a_1}^\dagger \\
& - \overline{q_{\alpha_1, a_2, i_1}} P_R u_{\alpha_1, i_2} Y_{u, i_1, i_2} \epsilon_{a_2, a_1} H_{a_1}^\dagger \\
& + H_{a_1} \mu^2 H_{a_1}^\dagger \\
& - \frac{1}{2} \lambda H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \overline{y_{1, i_1, r_1}} N_{r_1}^T C P_L \ell_{a_1, i_1} \text{NewScalar1}_{a_1}^\dagger \\
& - M_{S_1}^2 \text{NewScalar1}_{a_1} \text{NewScalar1}_{a_1}^\dagger \\
& + \text{lambdaH1Inv1} H_{a_3} \text{NewScalar1}_{a_4} \text{T3HiggsPortal1CGInv1}_{a_1, a_3, a_2, a_4} H_{a_1}^\dagger \text{NewScalar1}_{a_2}^\dagger \\
& + \text{lambdaH1Inv2} H_{a_3} \text{NewScalar1}_{a_4} \text{T3HiggsPortal1CGInv2}_{a_1, a_3, a_2, a_4} H_{a_1}^\dagger \text{NewScalar1}_{a_2}^\dagger \\
& + \frac{1}{2} \text{lambdaS1Inv1NewScalar1}_{a_3} \text{NewScalar1}_{a_4} \text{T3Scalar1SelfCGInv1}_{a_1, a_3, a_2, a_4} \text{NewScalar1}_{a_1}^\dagger \text{NewScalar1}_{a_2}^\dagger \\
& + \overline{\lambda_{T3}} \text{NewScalar2}_{a_1} \epsilon_{a_2, a_3} H_{a_1}^\dagger H_{a_2}^\dagger \text{NewScalar1}_{a_3}^\dagger \\
& - M_{S_2}^2 \text{NewScalar2}_{a_1} \text{NewScalar2}_{a_1}^\dagger \\
& + \overline{\ell_{a_2, i_1}} P_R N_{r_1} y_{2, i_1, r_1} \epsilon_{a_2, a_1} \text{NewScalar2}_{a_1}^\dagger \\
& + H_{a_1} H_{a_2} \text{NewScalar1}_{a_3} \lambda_{T3} \bar{e}_{a_2, a_3} \text{NewScalar2}_{a_1}^\dagger \\
& + \text{lambdaH2Inv1} H_{a_3} \text{NewScalar2}_{a_4} \text{T3HiggsPortal2CGInv1}_{a_1, a_3, a_2, a_4} H_{a_1}^\dagger \text{NewScalar2}_{a_2}^\dagger \\
& + \text{lambdaH2Inv2} H_{a_3} \text{NewScalar2}_{a_4} \text{T3HiggsPortal2CGInv2}_{a_1, a_3, a_2, a_4} H_{a_1}^\dagger \text{NewScalar2}_{a_2}^\dagger \\
& + \text{lambda12Inv1NewScalar1}_{a_3} \text{NewScalar2}_{a_4} \text{T3CrossScalarCGInv1}_{a_1, a_3, a_2, a_4} \text{NewScalar1}_{a_1}^\dagger \text{NewScalar2}_{a_2}^\dagger \\
& + \text{lambda12Inv2NewScalar1}_{a_3} \text{NewScalar2}_{a_4} \text{T3CrossScalarCGInv2}_{a_1, a_3, a_2, a_4} \text{NewScalar1}_{a_1}^\dagger \text{NewScalar2}_{a_2}^\dagger \\
& + \frac{1}{2} \text{lambdaS2Inv1NewScalar2}_{a_3} \text{NewScalar2}_{a_4} \text{T3Scalar2SelfCGInv1}_{a_1, a_3, a_2, a_4} \text{NewScalar2}_{a_1}^\dagger \text{NewScalar2}_{a_2}^\dagger
\end{aligned}$$

$$\begin{aligned}
& +D_{\mu_1}(H_{a_1})^\dagger D_{\mu_1}(H_{a_1}) \\
& +D_{\mu_1}(\text{NewScalar1}_{a_1})^\dagger D_{\mu_1}(\text{NewScalar1}_{a_1}) \\
& +D_{\mu_1}(\text{NewScalar2}_{a_1})^\dagger D_{\mu_1}(\text{NewScalar2}_{a_1})
\end{aligned}$$

$$\begin{aligned}
& \text{Matched EFT Lagrangian} - \frac{\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^4}{4(M_F^2 - M_{S1}^2)^2} \\
& - \frac{\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^4}{4(M_F^2 - M_{S2}^2)^2} \\
& - \frac{\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^4}{4(M_F^2 - M_{S1}^2)^2} \\
& - \frac{\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^4}{4(M_F^2 - M_{S2}^2)^2} \\
& - \frac{\hbar \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_3,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_4} M_{S2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F^3}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& + \frac{\hbar \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_3,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_4} M_{S1}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F^3}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& - \frac{\hbar \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_4,a_2} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& + \frac{\hbar \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S1}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_4,a_2} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& + \frac{\hbar y_{1,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S1}^2)} \\
& + \frac{\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S1}^2)} \\
& + \frac{\hbar y_{1,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4(M_F^2 - M_{S1}^2)^2} \\
& + \frac{\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S1}^2)^2} \\
& + \frac{\hbar y_{2,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S2}^2)} \\
& + \frac{\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S2}^2)} \\
& + \frac{\hbar y_{2,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4(M_F^2 - M_{S2}^2)^2}
\end{aligned}$$

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$$\begin{aligned}
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_1}^2)} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_1}^2)} \\
& + \frac{\hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{4 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_2}^2)} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_2}^2)} \\
& + \frac{\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{4 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_2}^2)^2} \\
& - \frac{\hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_3,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_4} M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F}{(M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{\hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_3,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_4} M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F}{(M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{\hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_1}^2 M_{S_2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_4,a_2} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{(M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{\hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_1}^2 M_{S_2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_4,a_2} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{(M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{\hbar B_{\mu_1 \mu_2}^2}{12 \epsilon} \\
& - \frac{1}{24} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2} \right) B_{\mu_1 \mu_2}^2 \\
& - \frac{1}{24} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2} \right) B_{\mu_1 \mu_2}^2 \\
& - \frac{(G_{\mu_1 \mu_2}^{X_1})^2}{4 g_s^2}
\end{aligned}$$

$$\begin{aligned}
& -\frac{\hbar \left(W_{\mu_1 \mu_2}^{A_1}\right)^2}{12 \epsilon} \\
& -\frac{1}{24} \hbar \log \left(\frac{\mu \text{bar} 2}{M_{S_1}^2}\right) \left(W_{\mu_1 \mu_2}^{A_1}\right)^2 \\
& -\frac{1}{24} \hbar \log \left(\frac{\mu \text{bar} 2}{M_{S_2}^2}\right) \left(W_{\mu_1 \mu_2}^{A_1}\right)^2 \\
& -\frac{\left(W_{\mu_1 \mu_2}^{A_1}\right)^2}{4 g_L^2} \\
& +i \overline{d_{\alpha_1, i_2}} \gamma^{\mu_1} P_R D_{\mu_1} \left(d_{\alpha_1, i_2}\right) \\
& +i \overline{e_{i_2}} \gamma^{\mu_1} P_R D_{\mu_1} \left(e_{i_2}\right) \\
& +i \overline{q_{\alpha_1, a_1, i_2}} \gamma^{\mu_1} P_L D_{\mu_1} \left(q_{\alpha_1, a_1, i_2}\right) \\
& +i \overline{u_{\alpha_1, i_2}} \gamma^{\mu_1} P_R D_{\mu_1} \left(u_{\alpha_1, i_2}\right) \\
& +i \overline{\ell_{a_1, i_2}} \gamma^{\mu_1} P_L D_{\mu_1} \left(\ell_{a_1, i_2}\right) \\
& -\overline{q_{\alpha_1, a_1, i_2}} P_R d_{\alpha_1, i_1} H_{a_1} Y_{d, i_2, i_1} \\
& -\overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} Y_{e, i_2, i_1} \\
& +\frac{\hbar y_{1, i_3, r_1} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{1, i_2, r_1} Y_{e, i_3, i_1}}{4 \epsilon} \\
& -\frac{1}{8} \hbar y_{1, i_3, r_1} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{1, i_2, r_1} Y_{e, i_3, i_1} \\
& -\frac{\hbar y_{1, i_3, r_1} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1, i_2, r_1} Y_{e, i_3, i_1}}{2 \left(M_F^2 - M_{S_1}^2\right)} \\
& -\frac{\hbar \log \left(\frac{\mu \text{bar} 2}{M_{S_1}^2}\right) \overline{y_{1, i_3, r_1} \ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1, i_2, r_1} Y_{e, i_3, i_1}}{2 \left(M_F^2 - M_{S_1}^2\right)} \\
& -\frac{\hbar y_{1, i_3, r_1} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^4 y_{1, i_2, r_1} Y_{e, i_3, i_1}}{4 \left(M_F^2 - M_{S_1}^2\right)^2} \\
& -\frac{\hbar \log \left(\frac{\mu \text{bar} 2}{M_{S_1}^2}\right) \overline{y_{1, i_3, r_1} \ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^4 y_{1, i_2, r_1} Y_{e, i_3, i_1}}{4 \left(M_F^2 - M_{S_1}^2\right)^2} \\
& +\frac{\hbar y_{2, i_3, r_1} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{2, i_2, r_1} Y_{e, i_3, i_1}}{4 \epsilon} \\
& -\frac{1}{8} \hbar y_{2, i_3, r_1} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{2, i_2, r_1} Y_{e, i_3, i_1} \\
& -\frac{\hbar y_{2, i_3, r_1} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2, i_2, r_1} Y_{e, i_3, i_1}}{2 \left(M_F^2 - M_{S_2}^2\right)} \\
& -\frac{\hbar \log \left(\frac{\mu \text{bar} 2}{M_{S_2}^2}\right) \overline{y_{2, i_3, r_1} \ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2, i_2, r_1} Y_{e, i_3, i_1}}{2 \left(M_F^2 - M_{S_2}^2\right)} \\
& -\frac{\hbar y_{2, i_3, r_1} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^4 y_{2, i_2, r_1} Y_{e, i_3, i_1}}{4 \left(M_F^2 - M_{S_2}^2\right)^2} \\
& -\frac{\hbar \log \left(\frac{\mu \text{bar} 2}{M_{S_2}^2}\right) \overline{y_{2, i_3, r_1} \ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^4 y_{2, i_2, r_1} Y_{e, i_3, i_1}}{4 \left(M_F^2 - M_{S_2}^2\right)^2}
\end{aligned}$$

$$\begin{aligned}
& -\overline{Y_{u,i_1,i_2}}\overline{u_{\alpha_1,i_2}} P_L q_{\alpha_1,a_1,i_1} H_{a_2} \bar{\epsilon}_{a_1,a_2} \\
& -\overline{Y_{d,i_1,i_2}}\overline{d_{\alpha_1,i_2}} P_L q_{\alpha_1,a_1,i_1} H_{a_1}^\dagger \\
& -\overline{Y_{e,i_1,i_2}}\overline{e_{i_2}} P_L \ell_{a_1,i_1} H_{a_1}^\dagger \\
& +C_{H2} H_{a_1} H_{a_1}^\dagger \\
& +\frac{\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon} \\
& -\frac{1}{8} \hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger \\
& -\frac{\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S1}^2)} \\
& -\frac{\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{S1}^2}\right) \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S1}^2)} \\
& -\frac{\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^4 y_{1,i_3,r_1} H_{a_1}^\dagger}{4(M_F^2 - M_{S1}^2)^2} \\
& -\frac{\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{S1}^2}\right) \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^4 y_{1,i_3,r_1} H_{a_1}^\dagger}{4(M_F^2 - M_{S1}^2)^2} \\
& +\frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon} \\
& -\frac{1}{8} \hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger \\
& -\frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S2}^2)} \\
& -\frac{\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{S2}^2}\right) \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S2}^2)} \\
& -\frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{4(M_F^2 - M_{S2}^2)^2} \\
& -\frac{\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{S2}^2}\right) \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{4(M_F^2 - M_{S2}^2)^2} \\
& -\overline{q_{\alpha_1,a_2,i_2}} P_R u_{\alpha_1,i_1} Y_{u,i_2,i_1} \epsilon_{a_2,a_1} H_{a_1}^\dagger \\
& -\frac{1}{2} \lambda H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& +\frac{\text{lambdaH1Inv}1^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& +\frac{5\text{lambdaH1Inv}2^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{6\epsilon} \\
& +\frac{\text{lambdaH2Inv}1^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& +\frac{5\text{lambdaH2Inv}2^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{6\epsilon} \\
& +\frac{\text{lambdaH1Inv}1\text{lambdaH1Inv}2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon}
\end{aligned}$$

$\sim$

$$\begin{aligned}
& + \frac{\text{lambdaH2Inv1}\text{lambdaH2Inv2}\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon} \\
& + \frac{1}{2}\text{lambdaH1Inv1}^2 \hbar \log\left(\frac{\mu_{\text{S1}}^{\text{bar}2}}{M_{\text{S1}}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{5}{6}\text{lambdaH1Inv2}^2 \hbar \log\left(\frac{\mu_{\text{S1}}^{\text{bar}2}}{M_{\text{S1}}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\text{lambdaH1Inv1}\text{lambdaH1Inv2}\hbar \log\left(\frac{\mu_{\text{S1}}^{\text{bar}2}}{M_{\text{S1}}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}} \\
& + \frac{1}{2}\text{lambdaH2Inv1}^2 \hbar \log\left(\frac{\mu_{\text{S2}}^{\text{bar}2}}{M_{\text{S2}}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{5}{6}\text{lambdaH2Inv2}^2 \hbar \log\left(\frac{\mu_{\text{S2}}^{\text{bar}2}}{M_{\text{S2}}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\text{lambdaH2Inv1}\text{lambdaH2Inv2}\hbar \log\left(\frac{\mu_{\text{S2}}^{\text{bar}2}}{M_{\text{S2}}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}} \\
& + \frac{\hbar \overline{\lambda_{\text{T3}}} H_{a_1} H_{a_2} \lambda_{\text{T3}} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\
& + \frac{\hbar \overline{\lambda_{\text{T3}}} H_{a_1} H_{a_2} M_{\text{S1}}^2 \lambda_{\text{T3}} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{\text{S1}}^2 - M_{\text{S2}}^2} \\
& + \frac{\hbar \log\left(\frac{\mu_{\text{S1}}^{\text{bar}2}}{M_{\text{S1}}^2}\right) \overline{\lambda_{\text{T3}}} H_{a_1} H_{a_2} M_{\text{S1}}^2 \lambda_{\text{T3}} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{\text{S1}}^2 - M_{\text{S2}}^2} \\
& - \frac{\hbar \overline{\lambda_{\text{T3}}} H_{a_1} H_{a_2} M_{\text{S2}}^2 \lambda_{\text{T3}} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{\text{S1}}^2 - M_{\text{S2}}^2} \\
& - \frac{\hbar \log\left(\frac{\mu_{\text{S2}}^{\text{bar}2}}{M_{\text{S2}}^2}\right) \overline{\lambda_{\text{T3}}} H_{a_1} H_{a_2} M_{\text{S2}}^2 \lambda_{\text{T3}} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{\text{S1}}^2 - M_{\text{S2}}^2} \\
& + D_{\mu_1} (H_{a_1})^\dagger D_{\mu_1} (H_{a_1}) \\
& - \frac{B_{\mu_1 \mu_2}^2}{4g_Y^2}
\end{aligned}$$
